

Recap

Data center sustainability is a serious issue for the future of the digital society

Non-functional requirements describe the quality of services and systems, that is, how they perform and the experience of their users

Quality refers to measurable properties:

Usability measures to what extent the technological infrastructure copes with the *characteristics* and *intensity* of the load it has to handle

Availability measures the *readiness* of a service to user requests

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

29

AWS outage: news coverage

The New York Times

Amazon Outage Forces Hundreds of Websites Offline for Hours

Amazon Web Services, a major provider of cloud services, cited a problem at its data center in Northern Virginia. The outage highlighted the fragility of global internet infrastructure.

- AWS outage affects major websites and apps globally
- Issue originated in US site known for outages
- Service improved, then problems recurred before recovery
- Gaming and financial platforms among those impacted by AWS issue

Amazon's AWS struggles to recover after major outage disrupts apps, services worldwide

Monday, 20th October 2025, 08:22 PM GMT+2

REUTERS

Amazon outage 'resolved' as Snapchat and banks among sites impacted

21 October 2025

Liv McMahon, technology reporter and Lily Jamali, North America

Share < Si

How a tiny bug spiraled into a massive outage that took down the internet

UPDATED OCT 21, 2025
By Lisa Enders and David Goldman

Amazon reveals cause of AWS outage that took everything from banks to smart beds offline

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

30

Cloudflare outage: news coverage

What is Cloudflare - and why did its outage take down so many websites?

Bet365, League of Legends, Sage, YouTube and Google were among those taken offline as provider tried to fix problem

● Cloudflare outage causes error messages across the internet

Robert Booth UK
technology editor

Tue 18 Nov 2025 16:23 CET

- Cloudflare was hit by an outage that knocked several major websites and services offline for several hours.

- OpenAI's ChatGPT, e-commerce platform Shopify and Elon Musk's social media platform X were among the sites impacted by the Cloudflare issues.

- A Cloudflare spokesperson said the company had identified the "root cause" of the outage, noting there's no evidence it was caused by an attack or malicious activity.

Internal server error Error code 500

Visit cloudflare.com for more information.
2025-11-18 11:35:32 UTC



You
Browser
Working



London
Cloudflare
Error



Host
Working

What happened?
There is an internal server error on Cloudflare's network.

What can I do?
Please try again in a few minutes.

Updated Nov. 18, 2025 9:35am ET

Cloudflare Stock Falls After It Reports Outages; X, ChatGPT Down for Many Users

By Joseph De Avila, Reporter

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

31

Cloudflare other outages: news coverage

Severe internet outages keep happening – and they might get worse

Growing reliance on a handful of major internet infrastructure companies has led to major disruptions, sparking everything from political pressure to computer science memes.

DOWN & OUT 'Half the internet' down AGAIN as another major Cloudflare outage leaves sites crippled

Cloudflare is one of the world's biggest network providers

Ethan Singh, News Reporter

Published: 09:16, 5 Dec 2025 | Updated: 10:51, 5 Dec 2025

Cloudflare admits 'we have let the Internet down again' after outage hits major web services - as it happened

Cloudflare reports it is investigating issues with Cloudflare Dashboard and related APIs

TECHNOLOGY · 6 min read · February 21, 2026

Cloudflare Outage Disrupts Major Websites Worldwide

A technical failure at Cloudflare on February 20 left millions unable to access Uber Eats, Bet365, and other popular platforms, exposing vulnerabilities in global internet infrastructure.

- ◆ Report widespread outages on February 20, 2026, affected major websites like Uber Eats, Bet365, Wikipedia, and Amazon Web Services due to Cloudflare's technical failure.

- ◆ Identify Cloudflare as the cause, impacting about 20% of all websites globally, highlighting its central role and single point of failure in internet infrastructure.

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

32

Crowdstrike outage

BBC

Mass IT outage affects airlines, hospitals, media and banks

19 July 2024

The New York Times

Chaos and Confusion: Tech Outage Causes Disruptions Worldwide

Airlines, hospitals and people's computers were affected after CrowdStrike, a cybersecurity company, sent out a flawed software update.

Global IT chaos persists as CrowdStrike boss admits outage could take time to fix

19 July 2024

Willkommen im Check-in 1

Abflug Departure

Handwritten note on clipboard:

S.No	Flight	Destination	STD	Gate
1.	UK 62	UDR	12:35	52
2.	UK 75	XR	13:15	52
3.	UK 77	AID	12:45	52
4.	UK 57	DCL	13:50	60
5.	UK 8	MAA	12:45	57-B
6.	UK 69	ATO	12:55	49
7.	UK 61	2X0	"	39-A
8.	UK 10	London Heathrow	13:35	39-B
9.	UK 5154	MAA	12:30	39-A
10.	UK 5162	RAI	13:25	56
11.	UK 5262	SXR	13:45	58
12.	UK 5204	UAN	13:45	48

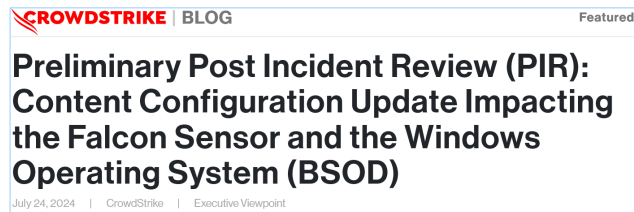
33

Outage impacts

34

What actually happened?

CrowdStrike released a preliminary post incident review a few days after the incident was detected



The problem was confined to Microsoft Windows systems, i.e., new Template Type deployed to detect attacks that abuse named pipes and other Windows IPC mechanisms

Mismatch between the 21 inputs validated by the Content Validator and the 20 provided to the Content Interpreter (see [Root Cause Analysis](#) for details)

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

35

Financial losses

Table 2: Fortune 500 Financial Loss by Industry

Industry Sector	Annual Revenue, All Companies	Annual Revenue, Impacted Companies	Estimated Direct Financial Loss	Estimated Direct Financial Loss, per Company
Health	\$2.77T	\$2.24T	\$1.94B	\$64.60M
Banking	\$0.82T	\$0.74T	\$1.15B	\$71.84M
Transportation - Airlines	\$0.19T	\$0.19T	\$0.86B	\$143.38M
Software & IT related Services	\$1.71T	\$0.89T	\$0.56B	\$112.59M
Retail/Wholesale	\$2.89T	\$0.73T	\$0.47B	\$14.73M
Other	\$5.15T	\$0.95T	\$0.19B	\$12.60M
Finance	\$0.68T	\$0.17T	\$0.14B	\$17.21M
Transportation - Other	\$0.36T	\$0.26T	\$0.07B	\$11.10M
Manufacturing	\$3.37T	\$0.31T	\$0.04B	\$5.93M
Total	\$17.94T	\$6.47T	\$5.41B	\$43.64M

Source: Parametrix Impact Analysis: CrowdStrike's Impact on the Fortune 500

GeoTech Cues | August 6, 2024

The Great IT Outage of 2024 is a wake-up call about digital public infrastructure

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

36

Five 9's metric for availability

Five-nines is the gold standard metric for availability, i.e., 99.999% uptime

This target requires robust infrastructure, extensive redundancy, automation and proactive monitoring

Significant increase in cost and complexity

Availability	Downtime/ Year	Downtime/ Month	Downtime/ Week	Downtime/ Day
99.999%	5.252 min	0.438 min	0.101 min	0.014 min
99.995%	26.28 min	2.19 min	0.505 min	0.072 min
99.990%	52.56 min	4.39 min	1.011 min	0.144 min
99.950%	4.38 hours	21.9 min	5.054 min	0.72 min
99.900%	8.76 hours	43.8 min	10.106 min	1.44 min
99.000%	98.6 hours	7.3 hours	101.077 min	14.4 min

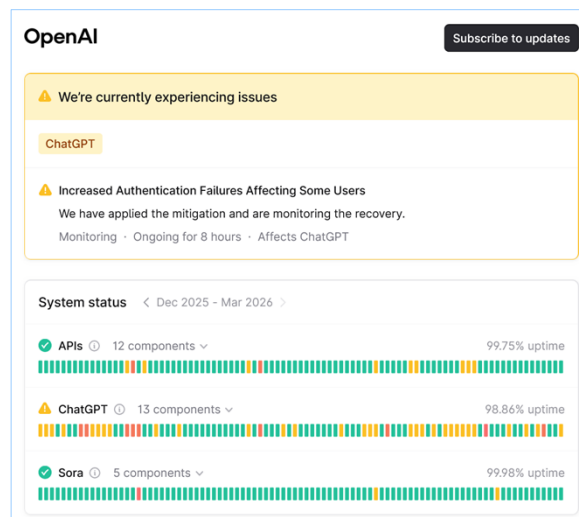
Organizations should carefully weigh the business value and impact of downtime against cost and complexity

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

37

OpenAI status

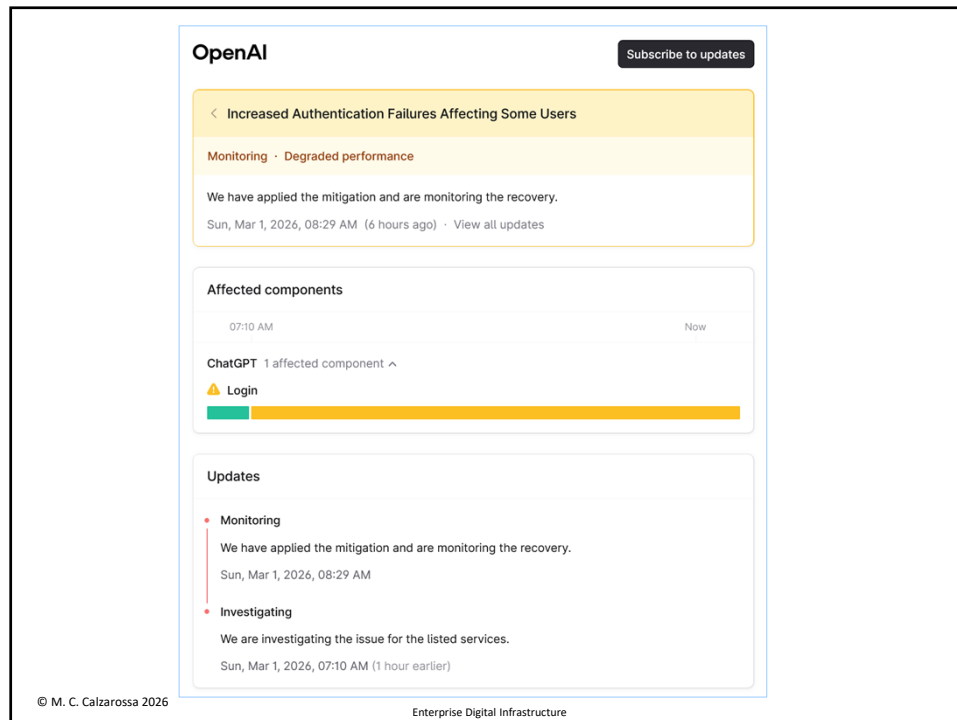


Availability metrics are reported at an aggregate level across all tiers, models and error types. Individual customer availability may vary depending on their subscription tier as well as the specific model and API features in use.

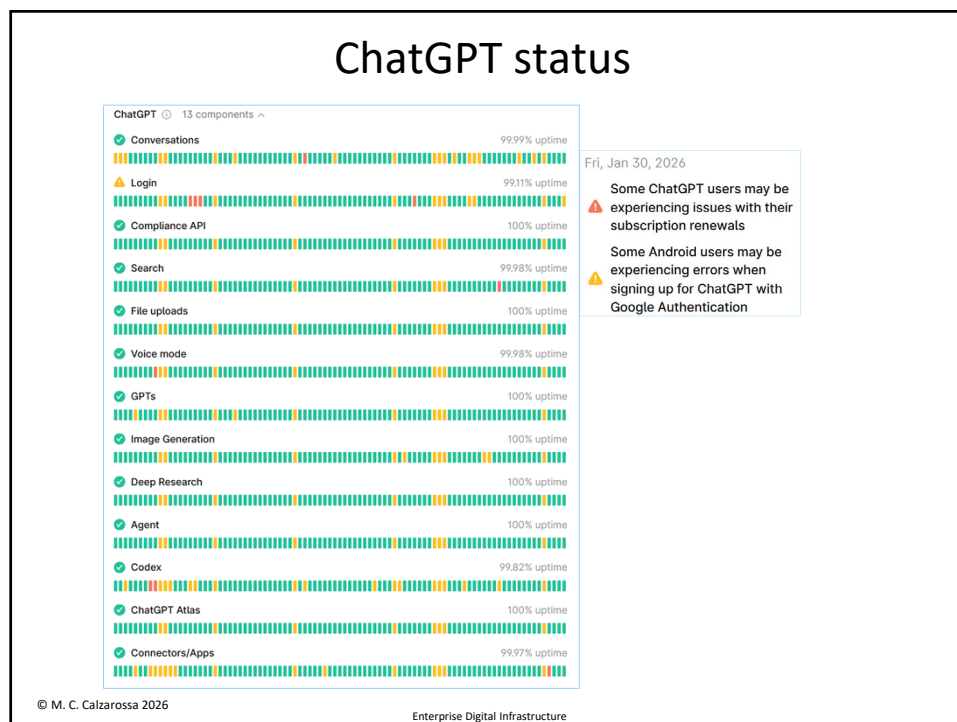
© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

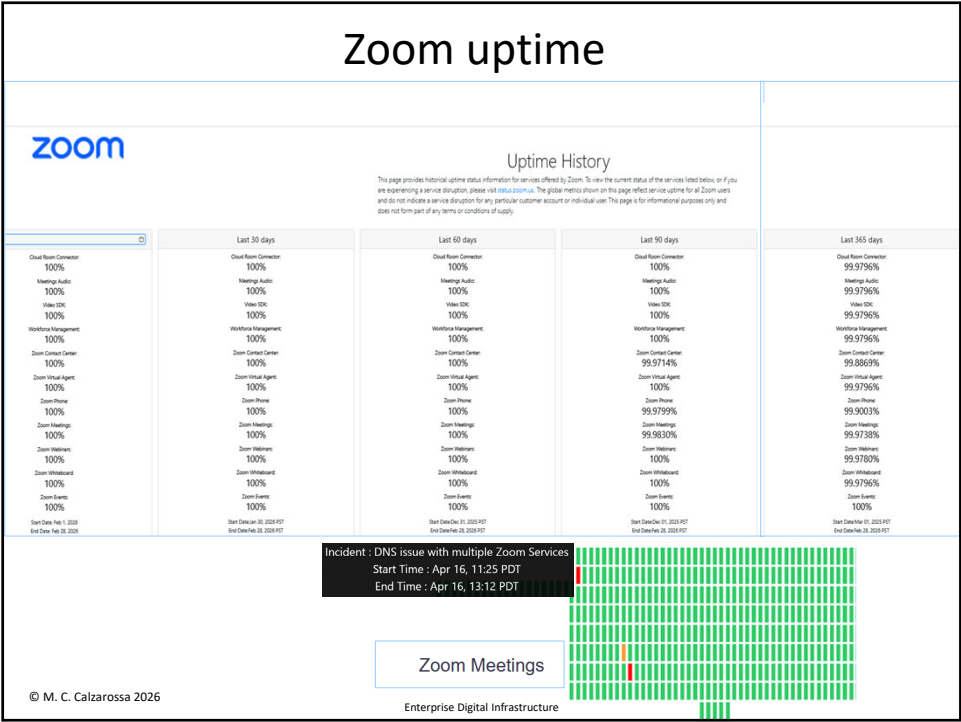
38



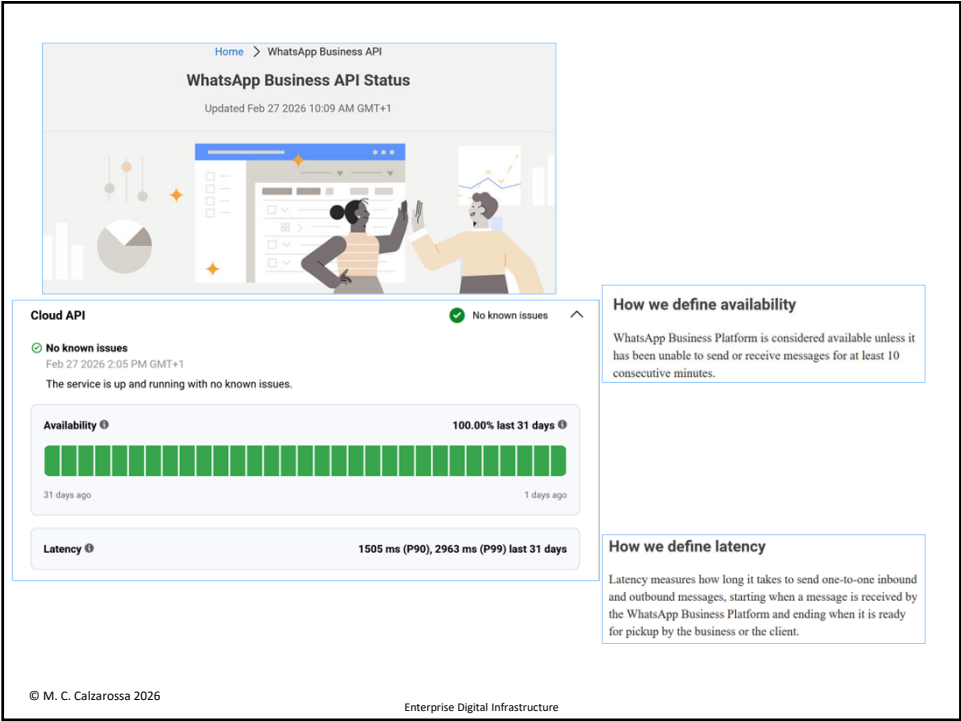
39



40



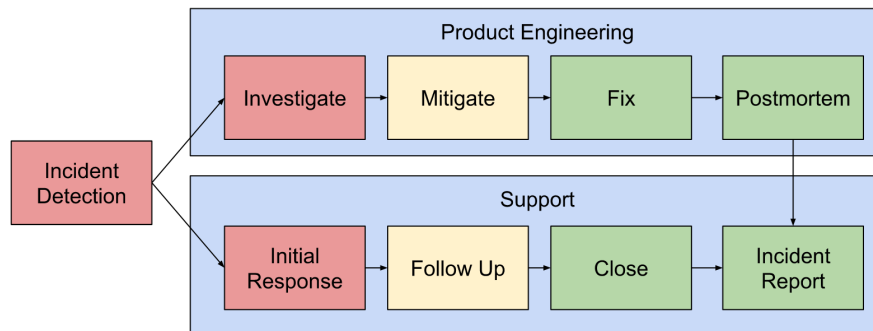
41



42

How Google manages incidents

Two parallel workflows each consisting of multiple steps



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

43

Scheduled downtimes

Scheduled downtimes refer to *planned* downtimes for system maintenance/upgrade



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

44

Maintenance announcement

Scheduled Maintenance



EOS Maintenance

Scheduled for Mar 5, 2026 07:00-10:00 UTC

To maintain the overall uptime and stability of Epic Online Services, we are performing scheduled maintenance on March 5, 2026 at 2:00 AM ET for the services listed below.

Impacted services:

There will be 1-2 hours of anticipated downtime for all Leaderboard functionality.

There will be up to 3 hours of degraded availability for the following services:

- Sessions:

- Existing sessions will time out and clients will be disconnected.
- New Invites may fail.
- Searching and matchmaking will be temporarily unavailable.

- Title Storage:

- Requests may fail and have to be retried.

- Player Data Storage:

- Requests may fail and have to be retried.

Once complete, the services above will automatically resume regular function and no action is required.

Posted on Feb 19, 2026 - 18:07 UTC

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

45

Outage analysis

Avoiding digital infrastructure unscheduled failures and downtimes is of paramount importance

According to [Uptime Institute](#), outages are becoming less frequent and less severe relative to the rapid growth of digital infrastructure

Cyber incidents are rising and often have severe, lasting impacts

Outage information is commercially sensitive and subject to “uncertainty”

Outages are costly in terms of financial loss, business and customer disruption and reputational loss (on average more than 100,000 USD)

Outages are mainly caused by:

- Failures of non-technological components (power issues)
- Human errors & management errors (ignored/inadequate procedures)
- Software, system & network errors (bugs, misconfigurations)
- **Third-party issues (cloud, services and telco providers)**

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

46

Availability: Lessons learnt

Downtimes are expensive and must be minimized (*pit stop*)

Unscheduled downtimes (due to power outage, software bugs, security attacks, misconfigurations, sudden traffic surge, ...) must be managed very carefully with timely announcements, regular follow-ups, mitigations and workarounds, detailed incident reports ... *information & transparency*

Necessary to *prevent* events that cause unscheduled downtimes

Scheduled downtimes (due to hardware/software maintenance, installation, configuration, upgrade, ...) must be planned very carefully with timely *announcements* and development of *mitigation plans* whenever possible

The availability depends on human errors and external events but also on the *reliability* of the technological infrastructure

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

47

Reliability

Reliability: probability of a hardware/software component to work as planned for a specified time interval (i.e., continuity)

Fault → *Error* → *Failure* chain

- *Fault*: defect inside a system, , e.g., software bug, that might cause an error
- *Error*: deviation of the system from the correct service state, e.g., software bug that manifests itself
- *Failure*: event that occurs when a system deviates from its correct behavior and stops performing the required function, e.g., software bug makes the system fail

A reliable component seldom fails

Components should be *robust* or *fault-tolerant* (i.e., able to mask faults)

© M. C. Calzarossa 2026

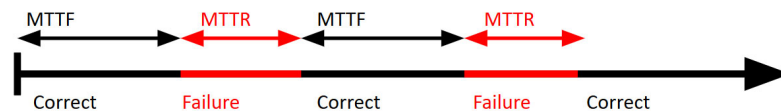
Enterprise Digital Infrastructure

48

Reliability (cont'd)

Robustness of a component (e.g., storage device) is measured in terms of MTTF (or MTBF or Annualized Failure Rate)

Recovery time of a failed component is measured in terms of MTTR



It is necessary to deploy specific *redundancies*

- *Replication* of hardware components
- *Diversity/distribution* of hardware and software components

Examples: power supplies, UPS, RAID architectures, providers

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

49

Reliability: Google's perspective

Google's application and network architecture is designed for **maximum reliability and uptime**. Google's computing platform assumes ongoing hardware failure, and it uses robust software failover to withstand disruption. All Google systems are inherently **redundant by design**, and each subsystem is not dependent on any particular physical or logical server for ongoing operation. Data is **replicated multiple times** across Google's clustered active servers so that, in the case of a machine failure, data will still be accessible through another system. We also replicate data to secondary data centers to ensure protection from data center failures.

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

50

Data center reliability

[Uptime Institute](#) created over 30 years ago a Tier classification and certification system to measure how well a data center can meet the needs of an organization

This proprietary standard explains the infrastructure required for data center operations

Each tier matches a particular business function and defines criteria for maintenance, power, cooling and fault capabilities

Four progressive Tiers: upper Tiers incorporate the requirements of the lower ones

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

51

Data center Tiers

Tier I: simple warehouse with an area for IT systems, engine generator, UPS & cooling system; shutdown required for preventive maintenance and repairs; severe consequences from failures

Tier II: partial redundancy with some redundant components for power and cooling

Tier III: “concurrently maintainable” with N+1 availability (Need plus 1); redundant distribution paths; no shutdown for component maintenance or replacement

Tier IV: 2N (Need times 2) redundancy and a 99.99% uptime guarantee, without any single point of failure; fault-tolerant and occurrence of any unplanned event while still maintaining operations

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

52

Reliability : Lesson learnt

Service continuity is ensured at the expense of an increased complexity (and cost) of the infrastructure

Trustworthiness

Trustworthiness (i.e., worth of trust/confidence) is the ability of a system to *protect* data, devices and infrastructures

This concept includes *security* and *data privacy*

Confidentiality, Integrity & Availability (CIA triad) of information is a common model used to develop security systems

Many types of security threats and attacks:

- DDoS
- data leakage/breach
- malware
- phishing
- ransomware
- ...

Denial of Service attacks

DDoS-ing is a crime

A Denial of Service (DoS) is a malicious targeted cyberattack whose goal is to take the “victim” (system or service) down and make it *unavailable* to its legitimate users



A DDoS, i.e., Distributed DoS attack, is a subclass of DoS attacks involving multiple devices, i.e., a **botnet**, to make attack stronger and more effective

DDoS attacks are popular “weapons” for diverse threat actors (geopolitical motivated hackers, cyber vandals) since these attacks are highly noticeable and impact an entire online user base

Attackers are willing to cause damages, productivity loss and financial losses or simply gain public attention

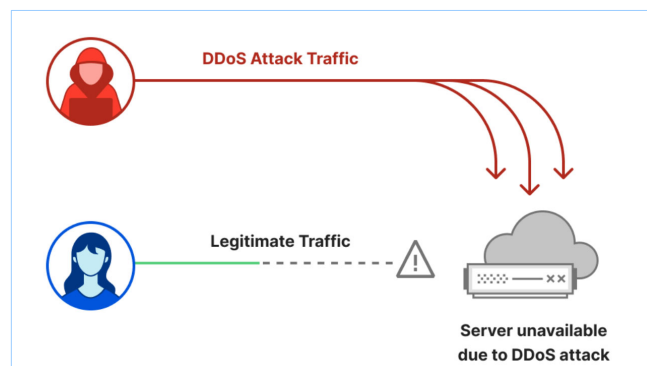
© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

55

Denial of Service attacks (cont'd)

Attacks are implemented by overloading the target with fake load with the objective of exhausting the target resources (e.g., bandwidth, processing capacity, memory)



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

56

Botnets

Botnet = robot + network: an army of comprised “zombie” devices injected with virus or malware and controlled from a remote location frequently without knowledge and consent of the owner

IoT devices (routers, home appliances) are often unmanaged & unmonitored, thus excellent targets of cyber-criminals for creating botnets (*thingbots*)

Devices infiltrated using different approaches, e.g., spam/phishing emails, brute force attacks, misconfigurations, exploitation of Common Vulnerabilities and Exposures (CVE)

Botnets can be created (building kits, tutorials for Building-Your-Own-Botnet) or rented for very little money

Botnets are ILLEGAL!!!

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

57

DDoS & botnets

A DDoS-for-hire is an *on demand* service powered by botnets consisting of two main components:

- Stresser, i.e., a software used to “stress-test” the resilience of a system
- Booter, i.e., the illegitimate use of a stresser

No need of technical know-how to launch attacks: any (low-skilled) individual can launch DDoS attacks by clicking a button

Affordable services available on the dark web and even on the web

DDoS-for-hire is ILLEGAL!!!

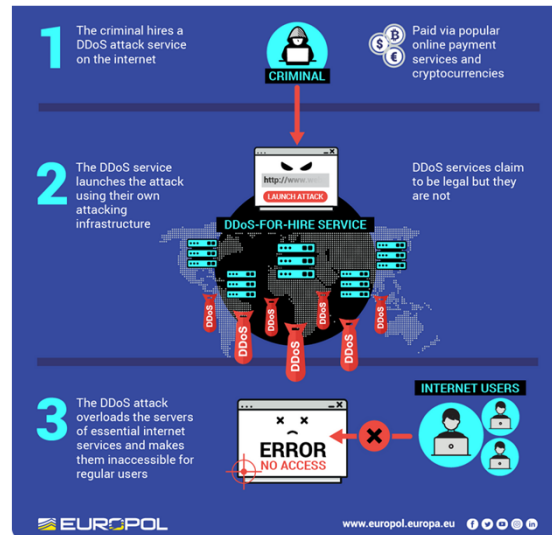
\$23.99 1 month	\$34.99 1 month	\$44.99 10 years
1 Month Gold	1 Month Diamond	Lifetime Bronze
Time per boot: 2400 sec	Time per boot: 3600 sec	Time per boot: 600 sec
Concurrents: 1	Concurrents: 2	Concurrents: 2
Total network: 220Gbps	Total network: 220Gbps	Total network: 220Gbps
Tools: Included	Tools: Included	Tools: Included
Support: 24/7	Support: 24/7	Support: 24/7
Buy with PayPal	Buy with PayPal	Buy with PayPal
bitcoin	bitcoin	bitcoin

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

58

How a DDoS attack can paralyze the Internet



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

59

Operation PowerOFF

Ongoing coordinated effort involving international law enforcement agencies, e.g., Europol Cyber Crime Center, aimed at dismantling criminal DDoS-for-hire infrastructures

Proactive steps to prevent future attacks (e.g., young people)

Operation PowerOFF: Law Enforcement Seizes 9 DDoS-for-Hire Webpages as Part of Global Crackdown

Operation PowerOFF Results in Four Arrests

By Sara Mosqueda | 8 May 2025 | Today in Security

Law enforcement shuts down 27 DDoS booters ahead of annual Christmas attacks

NCA infiltrates world's most prolific DDoS-for-hire service



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

60

Police from 15 Countries Shut Down Major DDoS-for-Hire Platforms



Vlad CONSTANTINESCU
December 12, 2024

Promo Protect all your devices, without slowing them down.
[Free 30-day trial](#)



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

61

Infection phase

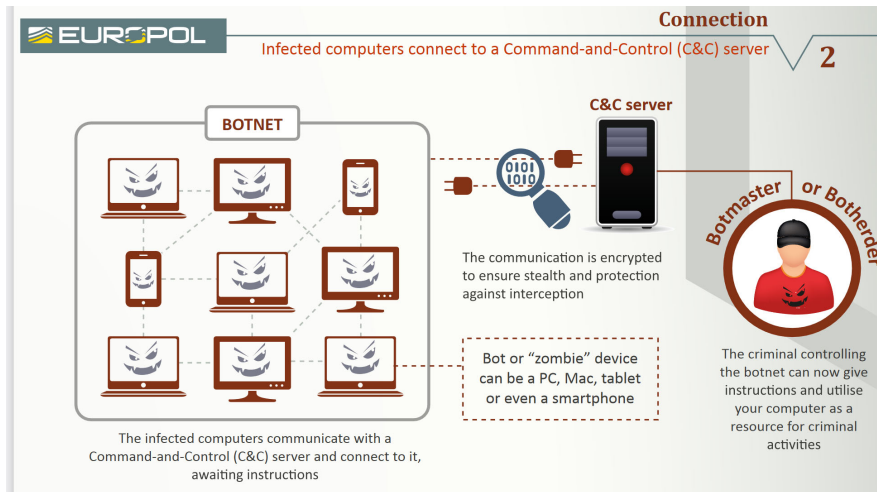


© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

62

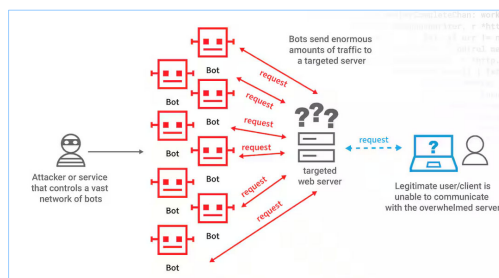
Connection phase



Enterprise Digital Infrastructure

63

DDoS implementation



The larger the botnet the more disruptive the attack (1-4 million devices)

Mirai botnet (first appeared in 2016) consists of IoT devices (e.g., DVRs, IP cameras, baby monitors) running on ARC processor and compromised by their exploiting default credentials

Aisuru/Kimwolf botnets (first appeared in 2024) consist of IoT devices (e.g., home routers, Android TV boxes) compromised by exploiting weak, default, or exposed credentials, unpatched firmware vulnerabilities

Enterprise Digital Infrastructure

64